

SHKODRA, Albania

WMO: 136000

Lat: 42.100N

Lon: 19.533E

Elev: 141

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.4	30.4	9.1	8.8	36.6	15.0	11.8	38.4	23.4	41.4	22.1	41.0	3.8	90	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.6	96.6	73.0	94.0	73.0	91.4	72.4	77.6	88.2	76.1	87.2	74.7	86.0	6.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	130.6	83.0	72.9	122.9	81.6	71.4	116.7	80.7	41.1	87.8	39.6	86.9	38.3	86.0	92.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.6	16.4	13.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			20.9	82.3	4.7	3.1	17.5	84.5	14.7	86.3	12.1	88.0	8.7	90.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.6	42.5	47.1	53.4	59.9	65.3	74.3	80.0	80.6	72.5	63.0	54.5	44.9	(6)
		DBStd	13.92	6.45	6.20	4.99	5.13	5.28	5.99	4.27	4.77	5.49	4.96	5.43	5.60	
		HDD50	570	240	111	24	2	0	0	0	0	0	0	19	174	
		HDD65	2825	697	501	359	164	61	4	0	1	6	95	315	622	
		CDD50	4804	8	30	130	300	476	730	931	949	675	403	155	17	
		CDD65	1583	0	0	0	12	72	284	466	485	230	33	1	0	
		CDH74	15493	0	2	2	97	438	2624	5109	5325	1751	145	0	0	
		CDH80	6860	0	0	0	10	69	1027	2473	2661	613	7	0	0	
(14)	Wind	WSAvg	5.5	5.0	6.1	6.4	5.2	5.3	5.1	5.1	6.0	5.2	5.8	5.1		(14)
(15)	Precipitation	PrecAvg	65.7	7.2	5.9	6.0	5.3	3.5	2.2	1.5	2.3	5.8	7.3	9.6	9.1	(15)
(16)		PrecMax	110.1	17.0	13.5	18.7	15.0	13.7	9.7	5.1	13.4	19.8	18.6	19.4	20.6	(16)
(17)		PrecMin	24.4	0.4	0.7	0.3	0.9	0.0	0.1	0.0	0.3	0.0	1.7	0.9		(17)
(18)		PrecStd	27.1	4.8	3.7	4.4	4.0	3.2	2.2	1.5	2.8	5.1	5.3	5.0	5.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	60.4	67.5	73.1	81.0	85.4	95.0	99.5	101.6	93.7	81.2	71.0	61.2		(19)
(20)		0.4% MCWB	54.5	53.9	54.6	63.7	68.2	73.9	74.4	72.8	71.6	66.7	62.6	56.4		(20)
(21)		2% DB	56.6	62.0	69.0	76.6	81.7	91.7	96.2	97.8	90.3	77.8	68.7	58.8		(21)
(22)		2% MCWB	51.7	53.6	54.8	60.8	66.2	73.2	73.7	72.6	70.9	64.2	61.7	53.6		(22)
(23)		5% DB	53.9	58.6	65.7	73.2	79.0	88.8	93.7	94.7	86.7	75.2	65.9	56.3		(23)
(24)		5% MCWB	49.2	52.2	53.0	58.9	65.6	71.8	73.6	72.2	69.2	63.4	58.5	50.4		(24)
(25)		10% DB	51.5	56.2	62.7	70.1	75.7	85.9	91.1	91.7	83.5	72.4	63.2	53.9		(25)
(26)		10% MCWB	47.4	50.2	52.3	58.1	64.2	71.2	72.6	72.2	68.7	63.1	57.1	49.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	55.7	58.4	59.5	65.9	71.2	78.8	79.9	79.5	76.5	70.5	65.4	58.0		(27)
(28)		0.4% MCDB	58.6	63.2	66.4	76.2	81.0	88.6	90.4	89.4	84.1	75.6	69.1	60.5		(28)
(29)		2% WB	52.6	55.3	57.6	63.8	68.9	76.5	77.5	77.2	74.3	67.9	62.4	55.3		(29)
(30)		2% MCDB	55.9	59.8	64.2	72.9	77.2	86.6	88.5	88.5	82.7	73.0	67.3	58.2		(30)
(31)		5% WB	50.3	53.3	55.9	61.6	67.3	74.7	76.0	75.7	72.7	66.2	60.4	52.1		(31)
(32)		5% MCDB	53.0	57.7	62.6	69.8	75.5	84.2	88.1	87.3	81.7	72.3	64.9	55.0		(32)
(33)		10% WB	48.2	51.1	54.2	59.7	65.7	72.8	74.5	74.4	71.1	64.5	58.3	49.6		(33)
(34)		10% MCDB	50.8	55.7	60.1	67.5	73.6	82.3	86.9	86.2	80.5	70.2	62.0	52.7		(34)
(35)	Mean Daily Temperature Range	MDBR	12.5	13.5	15.6	16.6	16.2	17.7	20.3	21.6	18.7	16.9	14.0	13.3		(35)
(36)		5% DB MCDBR	14.8	18.0	21.9	22.3	20.9	20.6	23.7	24.7	22.9	21.0	16.9	15.6		(36)
(37)		5% DB MCWBR	10.3	10.8	10.5	10.5	9.3	8.4	9.6	8.5	9.0	10.8	10.7	10.5		(37)
(38)		5% WB MCDBR	12.2	15.6	17.7	18.6	18.0	18.7	21.0	21.5	19.5	16.7	13.4	10.8		(38)
(39)		5% WB MCWBR	9.8	10.9	11.0	10.4	9.9	9.3	10.5	9.9	10.4	10.9	10.2	9.4		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.342	0.368	0.417	0.472	0.461	0.439	0.441	0.456	0.435	0.401	0.370	0.343		(40)
(41)		taud	2.398	2.305	2.186	2.056	2.156	2.271	2.259	2.209	2.256	2.326	2.362	2.401		(41)
(42)		Ebn at Noon	251	261	260	254	260	266	264	256	252	246	237	239		(42)
(43)		Edh at Noon	26	33	42	51	47	42	42	43	38	32	27	24		(43)
(44)	All-Sky Solar Radiation	RadAvg	529	768	1134	1474	1843	2135	2208	1962	1405	975	615	470		(44)
(45)		RadStd	61	115	156	163	147	142	116	125	115	101	64	86		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-185	N/A	N/A

Nomenclature: See separate page